

Linearni tipi

Vida Mlinar

Klasična logika

$$\frac{\frac{\Gamma \vdash P \quad \Gamma \vdash Q}{\Gamma \vdash P \wedge Q} \wedge\text{-intro}}{\frac{\Gamma \vdash P \wedge Q}{\Gamma \vdash P} \wedge_1\text{-elim} \quad \frac{G \vdash P \wedge Q}{\Gamma \vdash Q} \wedge_2\text{-elim}}$$

$$\frac{\Gamma, P \vdash Q}{\Gamma \vdash P \Rightarrow Q} \Rightarrow\text{-intro} \quad \frac{\Gamma \vdash P \Rightarrow Q \quad \Gamma \vdash P}{\Gamma \vdash Q} \Rightarrow\text{-elim}$$

$$\frac{P \in \Gamma}{\Gamma \vdash P}$$

$$\frac{\frac{P \in P}{P \vdash P} \quad \frac{P \in P}{P \vdash P}}{P \vdash P \wedge P}$$

Kako se naredi kill()?

Linearna logika

$$\frac{\frac{\Gamma \vdash P \quad \Delta \vdash Q}{\Gamma, \Delta \vdash P \wedge Q} \wedge\text{-intro}}{\frac{\Gamma \vdash P \wedge Q}{\Gamma \vdash P} \wedge_1\text{-elim} \quad \frac{\Gamma \vdash P \wedge Q}{\Gamma \vdash Q} \wedge_2\text{-elim}}$$

$$\frac{\Gamma, P \vdash Q}{\Gamma \vdash P \Rightarrow Q} \Rightarrow\text{-intro} \quad \frac{\Gamma \vdash P \Rightarrow Q \quad \Delta \vdash P}{\Gamma \vdash Q} \Rightarrow\text{-elim}$$

$$\frac{P \in \Gamma}{\Gamma \vdash P}$$

Klasična logika:

$$\frac{\frac{P \in P}{P \vdash P} \quad \frac{P \in P}{P \vdash P}}{P \vdash P \wedge P}$$

Linearna logika:

$$\frac{\frac{P \in P}{P \vdash P} \quad \frac{P \in P}{P \vdash P}}{P, P \vdash P \wedge P}$$

$$\frac{x : A \in \Gamma}{\Gamma \vdash x : A}$$

$$\frac{\Gamma \vdash M_1 : A_1 \quad \Gamma \vdash M_2 : A_2}{\Gamma \vdash (M_1, M_2) : A_1 \times A_2}$$

$$\frac{\Gamma \vdash M : A_1 \times A_2}{\Gamma \vdash fst(M) : A_1} \quad \frac{\Gamma \vdash M : A_1 \times A_2}{\Gamma \vdash snd(M) : A_2}$$

$$\frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash (x \mapsto M) : A \rightarrow B}$$

$$\frac{\Gamma M_1 : A \rightarrow B \quad \Gamma \vdash M_2 : A}{\Gamma \vdash M_1 M_2 : B}$$